



8-bit subtractor

Documented by:

Mohamed Sherif 2023/03849

Abdulkader Adnan 2023/07019

1. Truth Tables:

1-Truth table of different combinations of 8-bit minus 8-bit:

	A		B		C		D		E		F		G		H		I		J		K		L		M
1	A7	A6	A5	A4	A3	A2	A1	A0	B7	B6	B5	B4	B3	B2	B1	B0	D7	D6	D5	D4	D3	D2	D1	D0	Cout
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	1	0	1
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	1	1
6	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	1	1	1	1	0	0	1
7	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	0	1	1	1
8	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1	1	1	1	0	1	0	1
9	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	1	1
10	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	1	1	1	0	0	0	1
11	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	1	1	1	0	1	1	1	1
12	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	1	1	0	1	1	0	1
13	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	0	1	0	1	1
14	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1	1	1	0	1	0	0	1
15	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1	1	1	1	0	0	1	1	1
65529	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	0	0	0	0	1	0	0	0
65530	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	0
65531	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	1	1	0	0
65532	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	1	0	1	0
65533	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	0	0	0	0	0	1	0	0	0
65534	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	0
65535	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	1	0	0
65536	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	0
65537	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0

2-truth table of half adder:

A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$S = A'B + AB' = A \oplus B, C = AB$$

3-truth table of full adder:

A	B	Cin	Cout	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

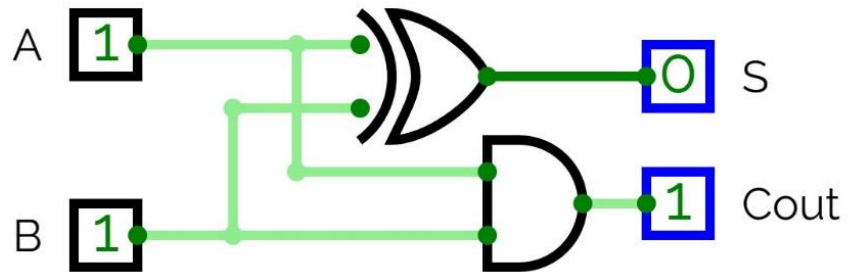
$$S = A'B'Cin + A'BCin' + AB'Cin' + ABCin = A \oplus B \oplus Cin$$

$$Cout = ACin + BCin + AB$$

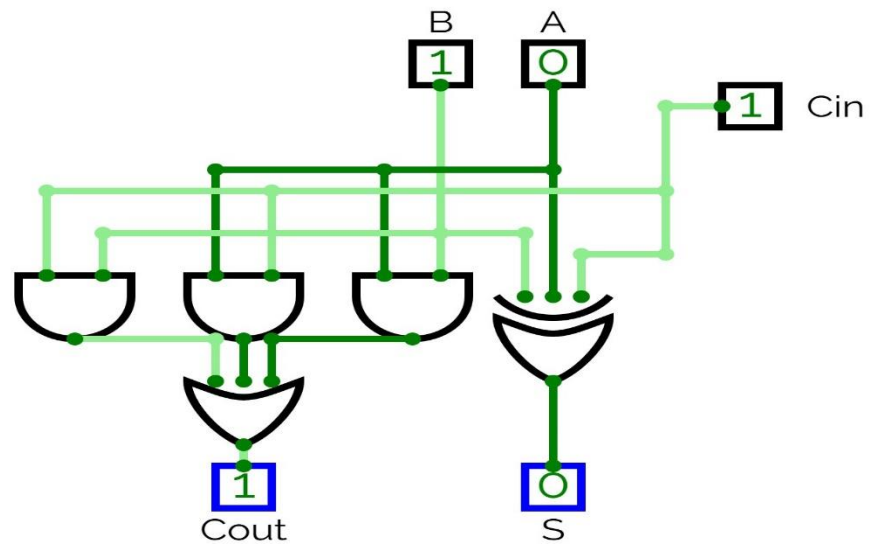
		B	
A	0	0	1
	0	1	1
		Cin	

2-Circuit Design with different INPUTS/OUTPUTS:

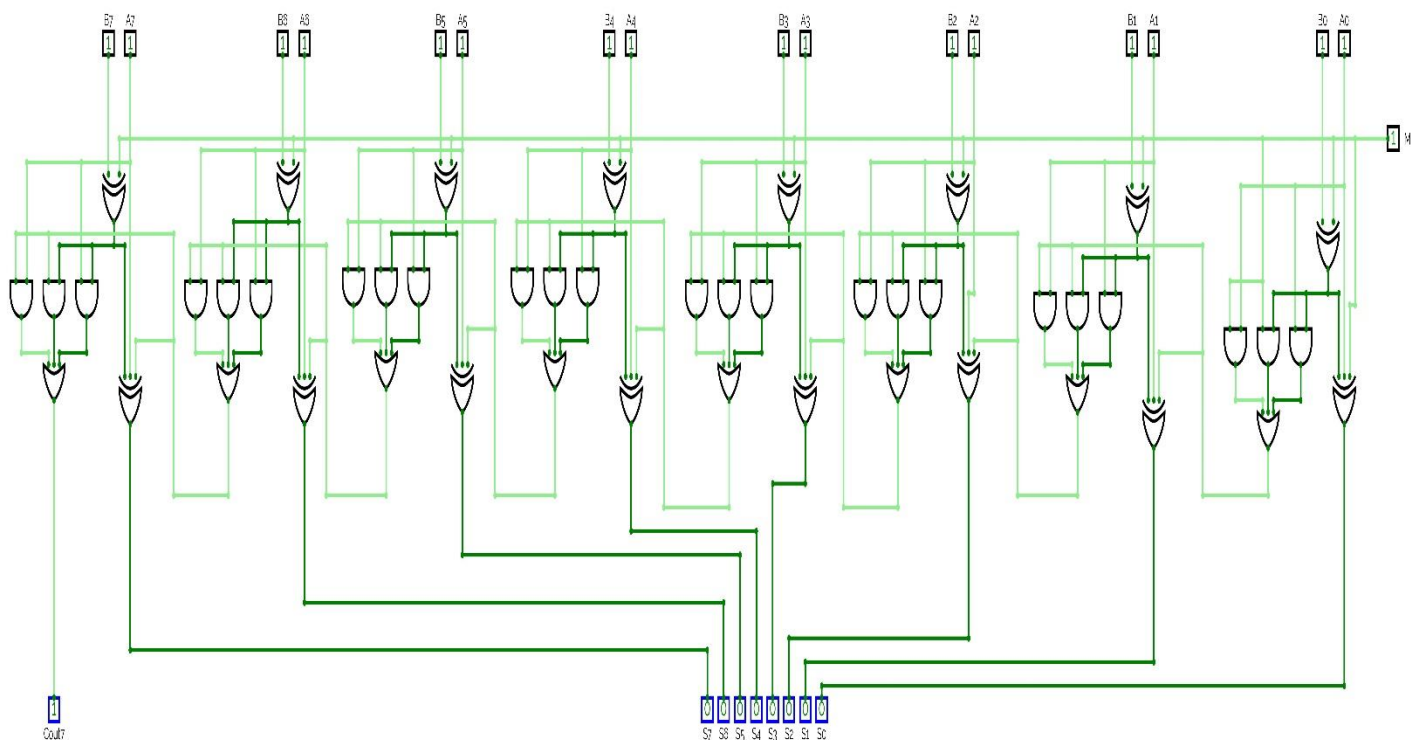
1- Circuit design of half adder:



2- Circuit design of full adder:



3- Circuit Design of 8-bit Adder-Subtractor:



3-How 8-bit subtractor does work?

To understand how 8-bit subtractor works, first we need to understand the subtraction using 2's complement:

say that we need to subtract B from A, ($A - B$) using 2's complement. First, we should get the 2's complement of B, by get the 1's complement of B and Adding one to B: Assuming that $A = 10101100$ & $B = 10101011$.

Step 1: get 1's complement of B, By converting every 1 to 0, and every 0 to 1.

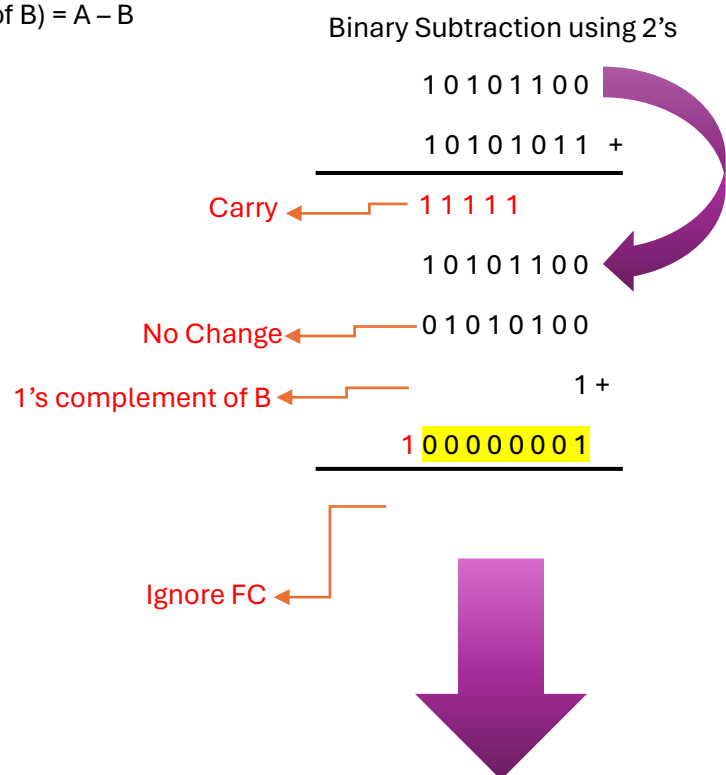
1's complement of B = 01010100

Step 2: Adding one to step 1 to get 2's complement of B :

2's complement of B = $01010100 + 1 = 01010101$

Step 3: perform the addition: $A + (2's \text{ complement of } B) = A - B$

Step 4: Ignore the final carry.



Same operation in Decimal:

172
171 -
001

4-The components of 8-bit adder/subtractor :

- 1- 8 full adders: each full adder consists of:
 - a. 1 XOR Gate with 3 inputs size.
 - b. 3 AND Gates.
 - c. 1 OR gate with 3 inputs size.
- 2- 8 XOR Gates can be replaced by 8 NOT gates.

It starts from the control switch (M). if $M = 0$ the circuit act as an 8-bit adder, bit if the $M = 1$ the circuit act as an 8-bit subtractor.

In case $M = 1$, The XOR gate Whose inputs are $M = 1$ and B act as Not gate. This way, we can apply the one's complement on B.

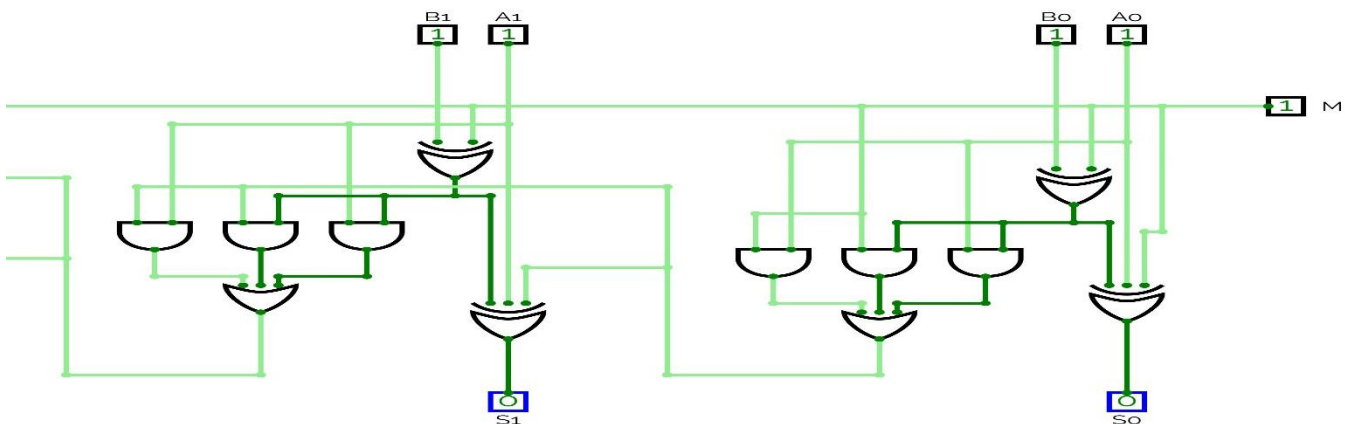
Logic function of XOR gate is: $F(A,B) = A'B + AB'$.

And in our case $A = M = 1$, $F(M=1,B) = (1)'B + (1)B' = (0)B + (1)B' = 0 + B' = B'$.

Then the complement of B' entre the next XOR gate and convert to 2's complement, due to $M = 1$.

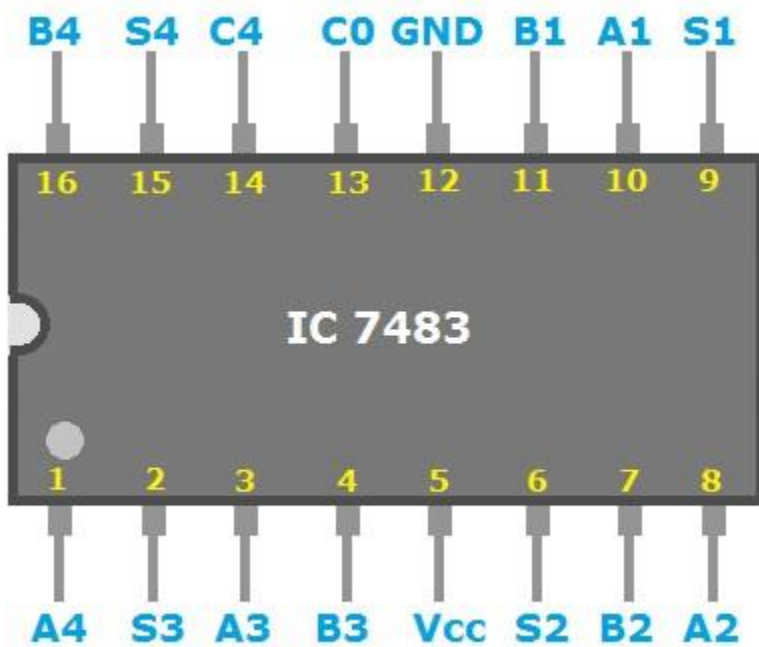
Where 2's complement of $B = B' + 1$. Then the subtraction is applied, because of Any number plus 2's complement of B means that This number minus B.

And for the carry we can implement the Boolean function of the carry: which is $Cout(A,B,Cin) = ACin + BCin + AB$. That means we need 3 AND gates & 1 OR gate with three inputs size. And the carry out should to the next full adder as well.



5-IC's Datasheet:

1. IC SN74LS83N



IC 7483 Pin Diagram

2. IC SN74HC86N

3.

